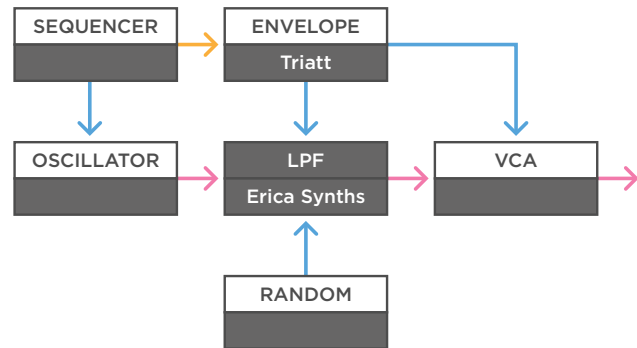
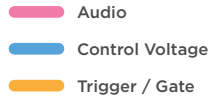


# Erica Synths Filters

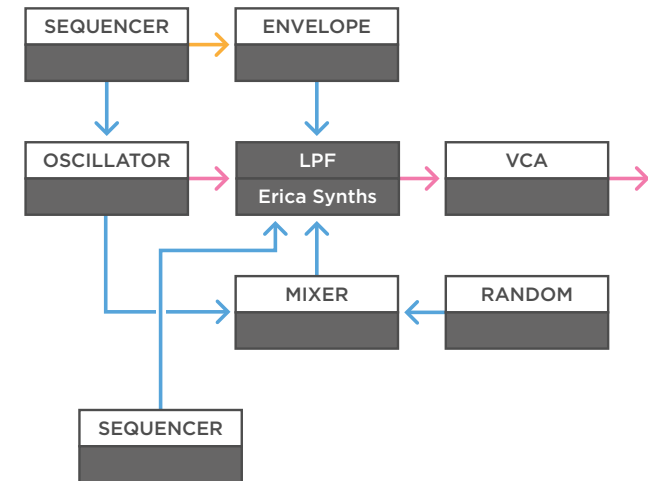
## Low Pass Filter



(a)

a - Here we have a simple melodic voice, created with an oscillator, filter and VCA, as well as a little sequencer to create a melody. We use an envelope generator, triggered by the sequencer, to shape the filter movement. We also add a somewhat tempo synced smooth random voltage to the filter to create some extra dynamics.

# MONOTRAIL TECH TALK

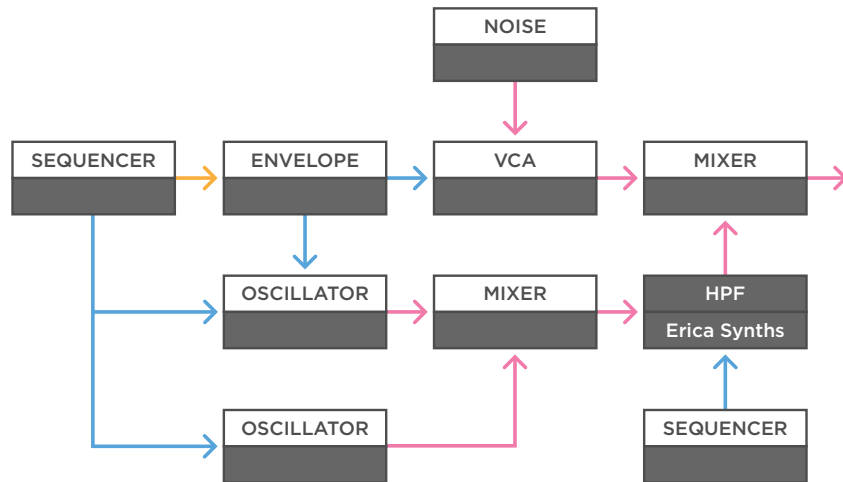
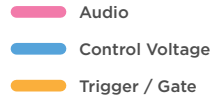


(b)

b - A nice trick to get more character out of a filter is to send it audio rate modulation. In our setup we can use any waveform from our main oscillator as our audio source and use another waveform from that same oscillator to then send it as a modulation source to the filter. In this case we use an external mixer, to mix our control voltages before going to the filter. We also add a second sequencer to the setup, so we can send a control voltage to the filter's resonance CV input. This works great to dial in steps with resonant accents.

# Erica Synths Filters

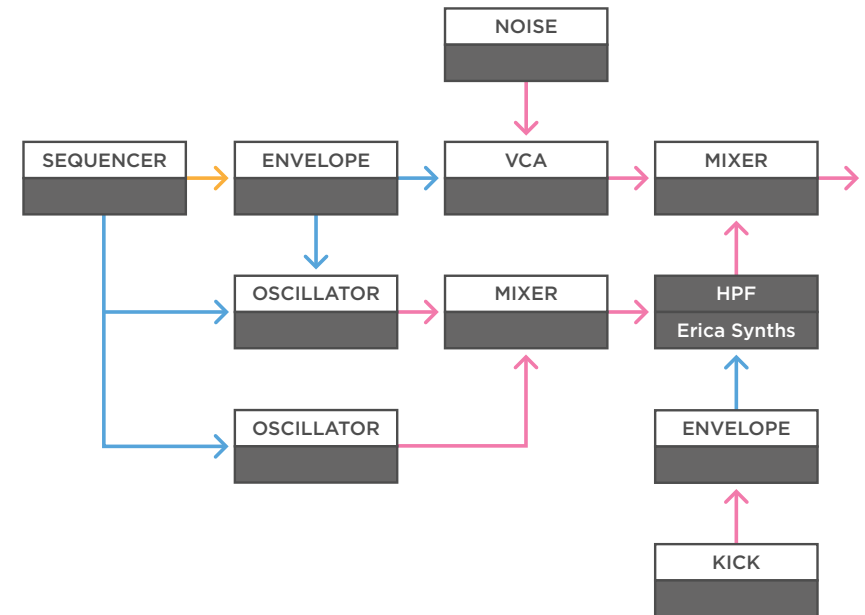
## High Pass Filter



(a)

a - In this setup we use two oscillators to create a percussive base sound. We mix the oscillators together and send them through a high pass filter. Then we use another mixer to mix in noise, which we send through a VCA first. A sequencer is used to control the pitch of the oscillators, and to trigger an envelope. The envelope is sent to the VCA to let through some noise, as well as the pitch on one of the oscillators, to create some transient. Then we use a second sequencer to control the cutoff frequency of the filter. This way we can exactly dial in the right amount of low end on each step. We can use this to create a nice balance between this sound, and a kick for example.

# MONOTRAIL TECH TALK



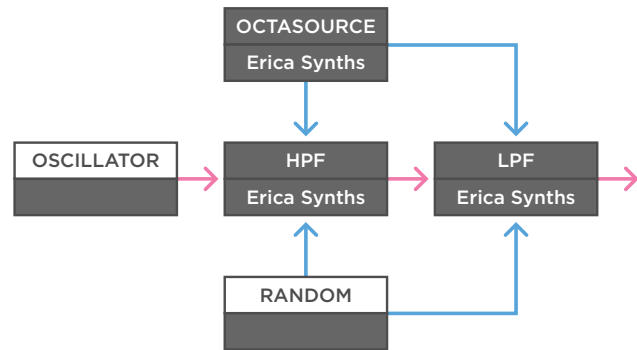
(b)

b - If you want a more direct side-chain effect in this patch, instead of the sequencer, you can use an envelope follower on something like a kick and send that to the filter. This way every time the kick, or another base heavy sound, hits, the filter cuts off more low end of your sound.

# Erica Synths Filters

## High Pass & Low Pass

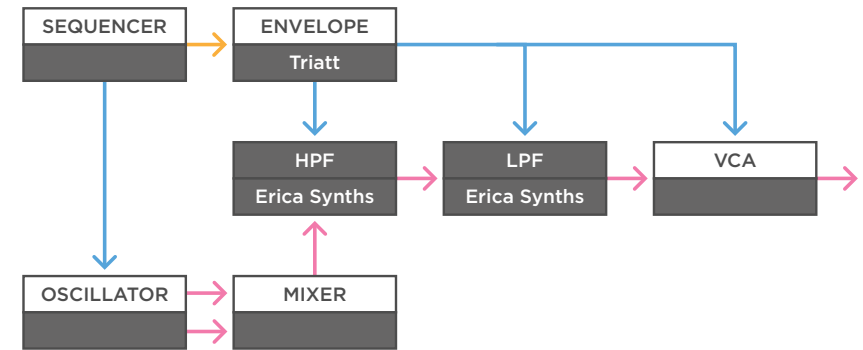
Audio  
Control Voltage  
Trigger / Gate



(a)

a - A classic setup is to run a high pass filter, into a low pass filter. This way you can use the high pass to accentuate low or mid frequencies, and then use the low pass filter to tame the harsh high end. And having two cutoff points, and resonance frequencies, you can really push a sound into the extreme. It's worth it to experiment with multiplying and inverting different modulation sources to both of the filters, to create extra movement.

# MONOTRAIL TECH TALK



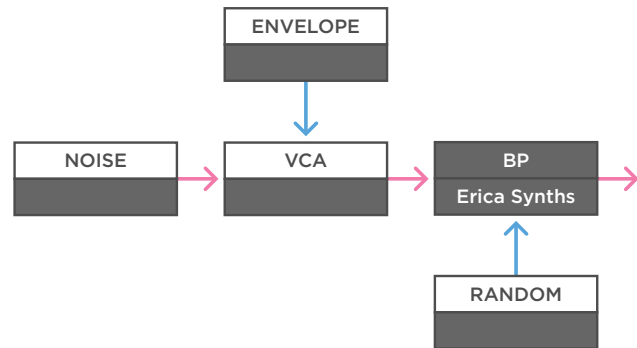
(b)

b - In this example we take the main and sub outputs from a wavetable oscillator to a mixer, and feed that in series through the High-pass and Low-pass filters, and finally, a VCA. We sequence the voice with a little sequencer that also triggers an envelope, modulating both filters, and the VCA.

# Erica Synths Filters

## Band Pass

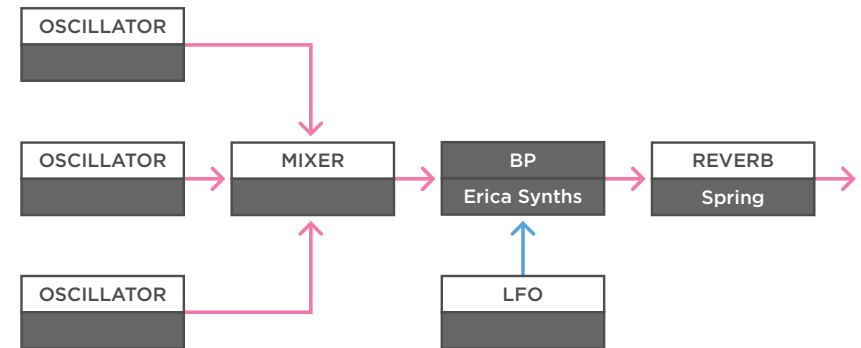
- Audio
- Control Voltage
- Trigger / Gate



(a)

a - A band pass filter is interesting to sculpt percussive sounds like a high-hat for example, made from noise through a VCA, that we send a steady envelope. With some smooth random modulation to the center frequency of the band pass filter, you can have the hat sound move around in the frequency spectrum.

# MONOTRAIL TECH TALK



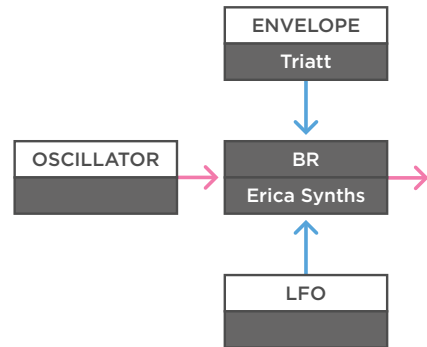
(b)

b - When we patch up a pretty huge drone sound with a few oscillators, we can use a band pass filter to make the sound thinner. When we send some modulation to the bandwidth control of the band pass filter, we can sweep a sound from very thin, to wide, and back again.

# Erica Synths Filters

## Band Reject

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - Band reject filters are also interesting for extreme sound sculpting. For example, let's take a wavetable oscillator again, and feed it to the band reject filter. We set the band pretty narrow and crank up the resonance on both the high pass and low pas filters. Then we send a simple sinewave LFO to the bandwidth of the filter, and a looping rising envelope to the center frequency.